UrbanLens — PRD Conformance Report

What the brief asked for, what the code actually does, and what is left. Audited against **prd.md** section by section on 15 August 2026, at commit **d54406e** (branch `fix/ui-backend-wiring-and-bugs`). SIH 2026 · PS-SW-001.

Every status below was checked against the source, an API response, or a measured run — not against intention. Where the implementation deliberately departs from the brief, it is marked **Substituted** and the reason is given, because a defensible substitution and an unmet requirement are different things and judges will ask which is which.

1. Headline

Area	Status	Evidence
The six required MVP features (§69)	**Built**	6 of 6 working end to end; `npm run demo` passes 12/12 steps on all four study areas
MVP priorities 1–12 (§68)	**Built**	12 of 12 present
API surface (§56)	**Built**	All 13 named routes implemented, plus 14 more
Explainable scoring (§18–20)	**Built**	Weights match the brief exactly; every result carries pros and cons
ML growth model (§11, §44)	**Partial**	XGBoost trained on real labels, ROC AUC 0.955–0.962 — but it scores development *pressure*, not a dated 2030 forecast
Satellite imagery (§31)	**Missing**	No Sentinel-2. NDVI/NDBI/NDWI never computed
PostGIS (§42)	**Substituted**	GeoJSON + shapely/pyproj/numpy; no database at all
Data-honesty rule (§30, §70)	**Caveat**	Mostly excellent — two layers breach it, see §6

One item to fix before you demo

Two toggleable map layers are labelled **“Vegetation & NDVI Canopy”** and **“Urban Heat Island (UHI)”**. NDVI and UHI are real satellite-derived indices, but these values come from a radial distance formula plus a character code from the cell id (`web/lib/mapdata.ts:155–182`). That is precisely what PRD §70 forbids — “fake GIS calculations… completely random scores” — and §30 forbids presenting synthetic data as real. Everything else in the product is scrupulous about provenance, which makes these two stand out. Remove them, or relabel as “illustrative — not satellite-derived”. Details in §6.

2. The six required MVP features (§69)

The brief says these six must be demonstrated *extremely well*. All six work.

#	Feature	Status	What actually happens
1	Interactive GLIS Map	**Built**	MapLibre GL; 15 toggleable layers; pan, zoom, click-to-select, hover tooltips, layer opacity, basemap switcher, Cmd+K search
2	Urban Growth + Prediction	**Built**	2018/2022/2026 timeline, built-up trajectory chart, corridor cards, 2030 probability grid from the trained model
3	Infrastructure Gap Analysis	**Built**	Per-ward scores across 5 services, population-weighted ranking, deficit heatmap, 15-minute analyzer
4	Smart Site Selection	**Built**	10 project types, constraint filtering, 6-factor weighted score, live weight sliders, ranked candidates
5	What-If Simulator	**Built**	Place an intervention, recompute coverage from the population raster, before/after comparison
6	AI Urban Planning Copilot	**Built**	7 tools over the real engine; Ollama for language, engine for every number

3. Module-by-module conformance

Core analysis

PRD §	Requirement	Status	Notes
§6	City Overview KPIs (12 listed)	**Built**	All 12 served by `/api/overview`; 6 surfaced on the panel, rest in Land/Infra panels
§7	Interactive GLIS map + 18 layers	**Partial**	15 layers built. Missing: true satellite-derived vegetation, water bodies as a layer, and a real land-use raster
§8	Parcel Intelligence Profile	**Built**	Every field in the brief's example exists, plus ranked recommended uses
§9	Urban Growth Analysis	**Partial**	Built-up change, growth %, direction and corridors — all modelled, not observed from imagery
§10	Urban Time Machine	**Built**	Year slider 2018->2026 redraws the built-up layer
§11	Growth Prediction (2030)	**Partial**	Probability grid with the brief's 5 risk bands. Scores present-day development pressure, not a dated forecast
§12	Expansion Corridors	**Built**	3–4 named corridors per area with growth, population, risk
§13	Infrastructure Gap Analysis	**Built**	Matches the brief's example output shape
§14	15-Minute City Analyzer	**Built**	8 facility types, per-point score. Straight-line distance at 4.8 km/h walk / 22 km/h drive
§15	Urban Livability Score	**Built**	7 weighted components, population-weighted city score

Land intelligence, decision support and AI

PRD §	Requirement	Status	Notes
§16	Smart Site Selection	**Built**	All 10 project types from the brief
§17	Parcel Ranking	**Built**	Ranked list, click to locate on map
§18	Urban Development Suitability	**Built**	Weights are exactly the brief's: 0.25 / 0.20 / 0.15 / 0.15 / 0.15 / 0.10
§19	Customisable planning weights	**Built**	Sliders re-rank live, client-side re-blend of engine factor scores
§20	Explainable recommendations	**Built**	Every candidate carries pros and cons, in the brief's tick / warning form
§21	Zoning Conflict Detection	**Built**	293 flagged for Ahmedabad — but official zoning is modelled, so these demonstrate the method, not real breaches
§22	Land-Use Change Detection	**Built**	Transitions pivoted from the parcel layer's history
§23	Vacant Government Land Finder	**Built**	Ownership + built-up + risk + access filters
§24	Land Opportunity Score	**Built**	Development-potential score per government parcel
§25	Environmental Constraints	**Partial**	Flood risk, water, vegetation modelled. No DEM, no real flood-zone data
§26	What-If Simulator	**Built**	7 intervention types
§27	Before vs After	**Built**	Coverage, distance, accessibility, livability, population covered
§28	AI Copilot	**Built**	Handles all 7 example questions from the brief
§29	LLM never computes spatial results	**Built**	Enforced structurally: the LLM picks a tool and phrases the result, the engine computes every figure. Without Ollama a deterministic router answers identically

4. Data sources (§30–37)

This is where the brief is most ambitious and the build is most partial. The project is unusually honest about it — every layer carries its own provenance on `/api/health` and `/api/layers`, and the weakest layer in play is what the UI reports.

PRD §	Source asked for	Status	What is actually used
§30	GLIS land records	**Substituted**	GLIS is not public. OpenStreetMap land polygons — real boundaries, real land-use tags — stand in. Labelled as such, never called cadastral
§31	Sentinel-2 + NDVI/NDBI/NDWI	**Missing**	No imagery pipeline. Built-up history for 2018/2022/2026 is modelled
§32	ISRO Bhuvan	**Missing**	Not integrated
§33	OpenStreetMap roads + POIs	**Built**	Overpass API. 1,252 facilities across 10 types, 2,480 road segments (Ahmedabad)
§34	Census of India + WorldPop	**Partial**	Census 2011 municipal totals projected to 2026, distributed by area × road density. No WorldPop raster
§35	GHSL historical built-up	**Missing**	This is the single blocker for a genuine temporal forecast
§36	Copernicus DEM	**Missing**	Elevation is modelled; it feeds flood risk
§37	AUDA planning data	**Missing**	Development-plan sheets are not published machine-readably; zoning is modelled
—	Ward boundaries	**Built**	Official — digitised municipal ward maps, 48 AMC + 11 GMC

5. Technology stack (§39–47)

PRD §	Asked for	Status	Reality
§39	Next.js, TypeScript, React, Tailwind, shadcn/ui	**Built**	Next 14.2.15, React 18.3, TS 5.6, Tailwind 3.4, Radix primitives
§40	MapLibre GL JS	**Built**	maplibre-gl 4.7
§40	deck.gl for large data	**Missing**	Not needed at current scale; MapLibre handles 2,567 parcels
§40	CesiumJS 3D (explicitly optional)	**Missing**	Brief says do not make it mandatory; 2D was prioritised as instructed
§41	Python, FastAPI, Pydantic, SQLAlchemy	**Partial**	FastAPI 0.135, Pydantic 2.12. No SQLAlchemy — there is no database
§42	PostgreSQL + PostGIS (“mandatory”)	**Substituted**	The largest single departure. Layers are GeoJSON; spatial work is shapely 2.1 + pyproj + numpy. See note below
§43	GeoPandas, Shapely, Rasterio, GDAL, pyproj	**Partial**	Shapely + pyproj only. GeoPandas/GDAL deliberately avoided so the install works without system GDAL
§44	scikit-learn, XGBoost	**Built**	XGBoost 3.4.1 classifier, scikit-learn 1.8 for split/CV/metrics
§45	OSRM / GraphHopper routing	**Substituted**	Brief permits “network-distance or drive-time approximations” for the prototype. Straight-line haversine at fixed speeds
§46	R2 / S3 object storage	**Missing**	No large binaries to store yet
§47	Vercel / Railway / Supabase deploy	**Missing**	Runs locally only
§48–55	9-table database schema	**Substituted**	No DB. The same entities exist as in-memory dataclasses and cached derivations

How to answer “where is your PostGIS?”

Do not pretend it is there. The honest answer is strong: *“Every ST_* operation the brief lists has a direct equivalent in what we use — ST_Contains is shapely's prepared predicate, ST_DWithin is our indexed raster window, ST_Distance is haversine. We index with an STRtree, which is the same R-tree PostGIS builds. At 2,567 parcels the whole city fits in memory and every query is sub-second, so a database would have added an ops dependency without changing a single number. It becomes necessary the moment we scale past one city — that is exactly where we would add it.”* Then show the scaling analysis in the technical reference.

6. Where the build breaks its own honesty rule

PRD §70 is unambiguous: mock datasets are acceptable, *fake analytics are not*. The project honours this almost everywhere — provenance per layer, confidence bands on thin data, explicit caveats on modelled zoning. Two things fall outside it.

Item	Severity	What is wrong	Fix
“Vegetation & NDVI Canopy” layer	High	Values are `0.2 + (radius/14)×0.6 + (charCodeAt(5)%10)×0.02`. NDVI is a specific satellite formula the brief spells out in §31. A judge who asks how it was computed will get an answer that contradicts the label.	Remove the layer, or rename to “Modelled green-cover gradient (illustrative)”
“Urban Heat Island (UHI)” layer	High	Values are `1.0 - (radius/12)×0.72` — a pure radial gradient with no thermal input.	Same: remove, or relabel as illustrative
Ownership / tenure	Low — already handled	Modelled for all but 10 of 2,567 parcels.	Already disclosed on /api/health and in the UI. No action
Official zoning	Low — already handled	Modelled, so zoning conflicts are method demonstrations.	Already disclosed in the conflicts response. No action

The bottom two rows are worth rehearsing as a *strength*: being able to say “this layer is modelled, here is the confidence, here is what would make it real” is more convincing to a technical judge than a dashboard that claims everything is authoritative.

7. What is left, in priority order

Ordered by demo value per hour of work.

#	Task	Effort	Why it matters
1	Remove or relabel the NDVI and UHI layers	15 min	Closes the only place the product overstates itself. Highest value per minute in the list
2	Deploy — Vercel + Railway/Cloud Run (§47)	2–4 h	A live URL is worth a great deal to judges and removes “it works on my machine” risk
3	Rehearse the §74 demo script end to end	1–2 h	The story is the product. `npm run demo` proves the data holds; the humans need the same rehearsal
4	GHSL built-up rasters for 2 dates (§35)	1–2 days	The one change that converts the growth model from “pressure” to a real dated forecast. Biggest technical credibility win available
5	Sentinel-2 NDVI/NDBI/NDWI (§31)	2–3 days	Would make §31 real and let the two flagged layers return honestly
6	PostGIS + vector tiles (§42)	1–2 weeks	Not needed for this demo; required for anything past one city
7	OSRM routing (§45)	1 day	Upgrades 15-minute analysis from straight-line to real drive time
8	Unit tests	ongoing	Two end-to-end harnesses exist; there is no unit layer

8. Verification as it stands

Check	Command	Result
PRD §74 demo story	npm run demo	12/12 steps, all four study areas
UI against real engine	npm run verify:ui	31 assertions, clean console and network
Types	npx tsc --noEmit	clean
Lint	npx next lint	clean
Production build	npm run build	succeeds
ML quality	/api/ml/model	ROC AUC 0.955–0.962, 5-fold CV 0.920–0.940

Prepared for the UrbanLens team · figures measured on the commit named on page 1 · statuses reflect code, not intent.